



SIMATS ENGINEERING
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AI-Matched University–Industry Capstone Collaboration

Platform with Skill-Gap Recommendation

CSA0925- JAVA PROGRAMMING

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SIMATS Engineering

Problem Background

❑ Application Domain

- Artificial Intelligence
- Education Technology (EdTech)
- Industry Collaboration
- Recommendation Systems

❑ Need for the Project

- Automate project allocation.
- Improve university–industry collaboration.
- Enhance student employability.
- Reduce faculty workload.

❑ Current Scenario

- Manual capstone project allocation.
- Limited industry participation.
- Skill mismatch between students and projects.
- Lack of personalized recommendations.

❑ Target Users

- Students
- Faculty
- Industry Mentors
- Administrators

Problem Statement

- **Problem Definition**

- Universities face challenges in assigning students to industry-relevant capstone projects due to manual selection processes and lack of skill-based matching.

- **Evidence**

- Manual allocation consumes significant faculty time.
- Many students lack required industry skills.
- Companies struggle to identify suitable students.

- **Limitations of Current Systems**

- Manual matching
- No AI recommendations
- No skill-gap analysis
- Poor collaboration

- **Expected Impact**

- Accurate project matching
- Improved employability
- Strong industry collaboration
- Better project success rate

Objectives

- To develop an AI-based platform for matching students with industry-sponsored capstone projects.
- To identify skill gaps between student profiles and project requirements.
- To recommend personalized learning resources for skill improvement.
- To increase students' employability through industry-relevant projects.

Project Scope

Develop a Java-based AI platform that matches students with industry capstone projects and recommends skill improvements through intelligent skill-gap analysis.

Literature Survey

Sl. No	Year of Publication	Authors	Title of the Publication	Findings	Limitations
1	2020	Gupta, P., Singh, R., and Sharma, A.	AI-Driven Capstone Project Matching System for University–Industry Collaboration	Proposed an AI-based matching approach using student profiles and project requirements to improve capstone allocation.	Does not consider skill-gap analysis and personalized learning recommendations.
2	2021	Patel, M. and Verma, S.	Skill Gap Analysis and Recommendation System Using Machine Learning	Identified skill gaps using machine learning and recommended courses to bridge the gap for better employability.	Limited to academic skill sets and not integrated with project matching.
3	2022	Reddy, K., Joshi, P., and Kumar, V.	Hybrid Recommendation System for Academic Project Allocation	Developed a hybrid filtering model (content-based + collaborative) for recommending suitable capstone projects.	Lacks continuous learning and real-time industry feedback integration.
4	2023	Sharma, A. and Chakraborty, D.	University–Industry Collaboration Platform Using Cloud and AI	Built a cloud-based platform enabling real-time collaboration, progress tracking, and communication between universities and industries.	No personalized skill enhancement recommendation mechanism.
5	2024	Khan, I., Bose, T., and Mehta, P.	AI-Powered Skill Recommendation and Career Path Prediction for Students	Used AI to predict suitable career paths and provide skill improvement recommendations based on industry trends.	Not integrated with capstone project matching and industry collaboration.

Existing System & Proposed System

Existing System	Proposed System
• Manual Matching	• AI-Based Matching
• Limited Industry Support	• Real-Time Collaboration
• No Skill Analysis	• AI Skill-Gap Analysis
• Static Recommendations	• Personalized Recommendations
• Manual Reports	• Automated Reports

Novelty: Integrated AI matching with skill-gap recommendation and collaboration.

Requirements

Functional Requirements

- Student Registration
- Industry Registration
- AI Matching
- Skill Analysis
- Report Generation

Non-Functional Requirements

- Security
- Scalability
- Reliability
- Performance
- User-Friendly Interface

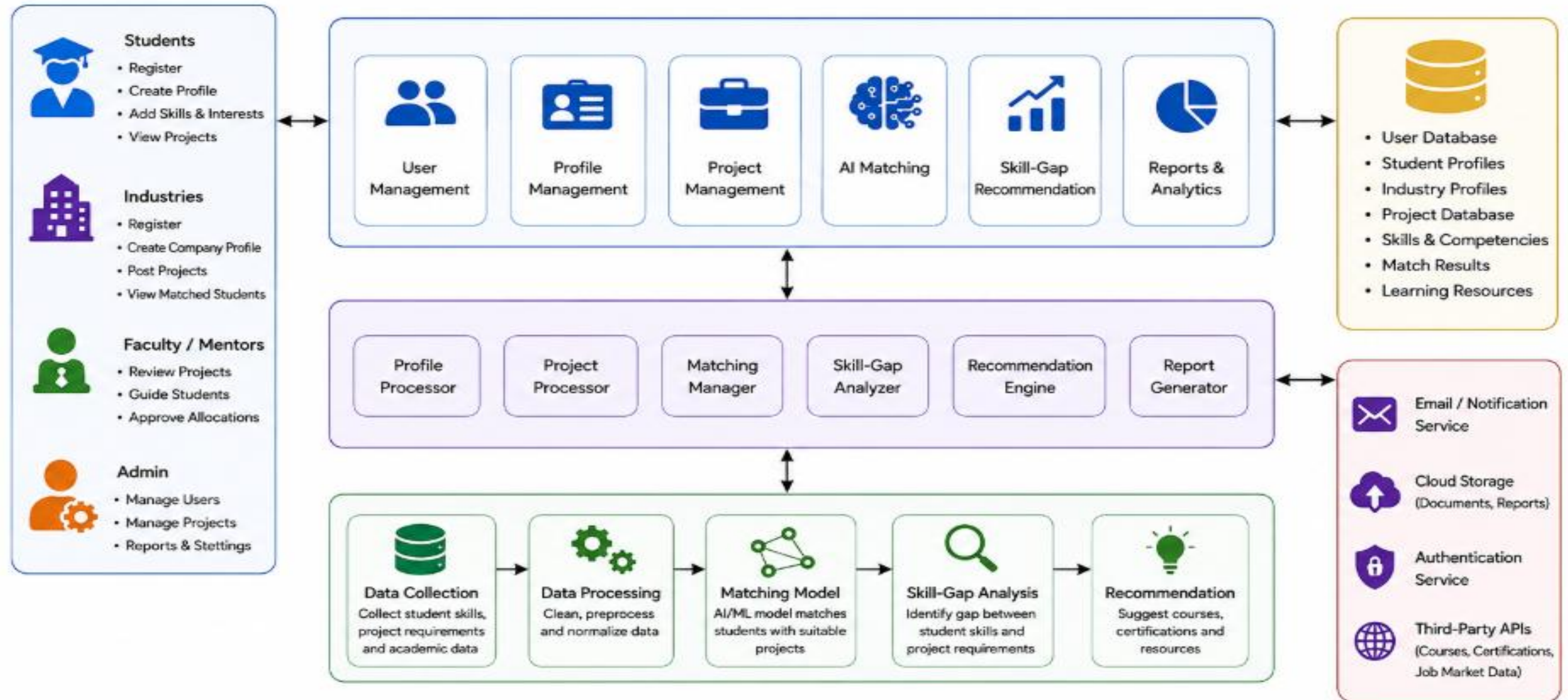
Dataset/APIs

- Student Skills Dataset
- Industry Project Dataset
- Course Recommendation API
- Job Skills Database

Hardware

- Intel i5 Processor or above
- 8 GB RAM
- 256 GB SSD

System Architecture



Project Planning

Work Breakdown Structure (WBS)

Requirement Analysis

System Design

Database Design

AI Model Development

Frontend Development

Backend Development

Testing

Deployment

Milestones

Requirement Completion

Design Completion

AI Model Completion

System Testing

Final Deployment

Expected outcomes

Deliverables

AI-Based Matching Platform

Skill-Gap Recommendation System

Student & Industry Portal

Reports & Analytics Dashboard

Benefits

Accurate project allocation

Better student–industry collaboration

Reduced manual effort

Enhanced employability

Improved project success rate

References :

I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*. MIT Press, 2016.

C. M. Bishop, *Pattern Recognition and Machine Learning*. Springer, 2006.

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S. Russell and P. Norvig, *Artificial Intelligence: A Modern Approach*, 4th ed., Pearson, 2021.

GPS Photo



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